

Literature Review: Machine Learning Approaches to FIFA World Cup 2026 Winner Prediction

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1 Introduction

Predicting outcomes in association football has attracted sustained academic interest because of the sport’s inherent stochasticity and global reach. The expansion of the 2026 FIFA World Cup to 48 teams, producing 104 matches across the United States, Canada, and Mexico, presents a richer and more complex prediction target than any prior edition. This project develops an end-to-end machine learning pipeline to estimate match-level outcome probabilities and simulate the 2026 FIFA World Cup to produce tournament winner probabilities. This literature review surveys the principal methods, datasets, and modeling frameworks that inform and motivate our approach.

2 Feature Engineering and Data Sources

Historical match data spanning more than a century of international football, with results considered from 1872 onward, forms the backbone of most predictive systems, and similarly serves as the primary data source in our project. The literature identifies several feature families that consistently improve predictive performance:

- **Elo ratings** — a dynamic, zero sum measure of team strength updated after each match. Elo scores compress squad quality, recent form, and competitive history into a single auditable format, making them a widely adopted baseline feature [2].
- **FIFA ratings and bookmaker odds** — market implied probabilities from bookmakers and official FIFA rankings provide complementary strength estimates grounded in both statistical history and collective expert judgement [1].
- **Player level attributes** — average squad ratings and transfermarkt market values aggregate individual player quality into team-level covariates, capturing depth and

star power.

- **Contextual covariates** — host advantage, climate similarity, average travel distance, and macroeconomic indicators (GDP) have been shown to add predictive signal, particularly in intercontinental tournaments where environmental factors vary significantly across teams.

In our project, we engineer a composite *geographic advantage score* as a weighted combination of host advantage, climate similarity, and travel distance—a design motivated by prior work demonstrating the amplified role of venue in multi-host tournaments [4].

3 Modelling Approaches

3.1 Gradient Boosting (XGBoost)

Our project uses an XGBoost classifier to estimate match-level outcome probabilities. These probabilities are then propagated through the 2026 tournament structure using Monte Carlo simulation to estimate overall winner probabilities. XGBoost is suitable for this task because it performs well on structured tabular data and can capture non-linear interactions between team strength, historical performance, and contextual features.

3.2 Hybrid Statistical Models

Beyond pure machine learning, the literature demonstrates the value of hybrid approaches. Zeileis et al. (2026) fuse a Poisson based goal scoring model, trained on all major men’s tournaments from 2006 to 2024, with bookmaker consensus and player market values. This combination of domain specific assumptions with data driven learning gives well calibrated forecasts capable of simulating full tournament brackets. Our pipeline draws on this philosophy by bringing in both statistical features and learned representations.

3.3 Probabilistic and Elo Based Baselines

Simpler Elo based baselines serve a dual purpose in the literature: providing a transparent benchmark and underpinning publicly available probability datasets [4]. We use such baselines in our project to validate our model against a principled lower bound before introducing more complex features.

4 Evaluation and Calibration

A recurring challenge in football prediction is model calibration. A classifier assigning 60% win probability to a team should be correct in roughly 60% of comparable cases. Prior

work shows that careful calibration rather than raw accuracy alone best distinguishes models in football contexts [3]. Tournament level evaluation requires simulating bracket outcomes through Monte Carlo methods, accumulating uncertainty across up to eight successive knockout rounds. Our project adopts this simulation based evaluation strategy to produce probabilistic winner forecasts rather than single point predictions.

5 Research Gap and Project Motivation

While the reviewed literature establishes strong foundations, several gaps motivate the present project. First, most existing pipelines were designed for the 32 team format and have not been systematically adapted to the structural changes introduced by the 2026 expansion to 48 teams, where the additional group stage matches and revised knockout bracket alter the probability landscape considerably. Second, contextual covariates such as geographic advantage and climate similarity are rarely combined with player level features in a single pipeline, leaving potential predictive signal unexploited. Third, few studies provide fully reproducible, end-to-end implementations that can be updated as the tournament progresses.

This project aims to address these gaps by constructing a machine learning pipeline that integrates historical match data, engineered contextual features, and gradient boosting within a format aware framework tailored specifically to the 2026 edition. The goal is to produce well calibrated, interpretable winner predictions that can serve as a foundation for further analysis as the tournament unfolds.

Bookmaker odds and rating-based probabilities have also been used as benchmark inputs for forecasting international football tournaments [4].

References

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